

Routes that detour

Pattern routers ignore blocked channels

The idea

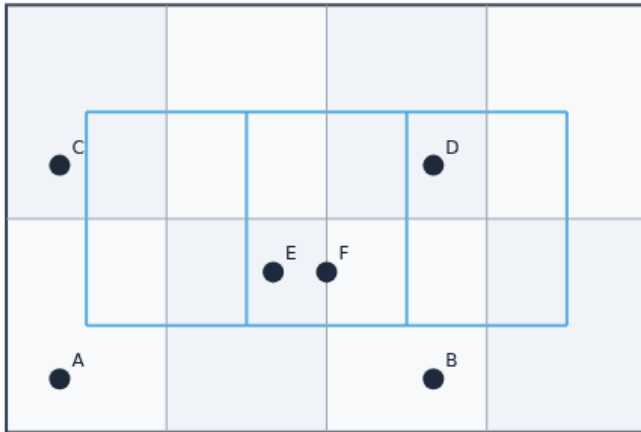
- Seed a queue with start and path list
- Pop, expand neighbors, skip edges with usage greater than or equal to capacity
- First time you dequeue the goal, return the path
- If the queue empties, return None
- Shortest feasible path wins

Edge cost

MAZE ROUTING

Edge cost

Step 1 / 5 · cost · module02-05-maze-gcell-route



Explanation

Maze adds penalty when usage $\geq$ capacity.

- Search on GCells
- Detour allowed

$$\text{Cost} = 1 + \text{penalty}$$

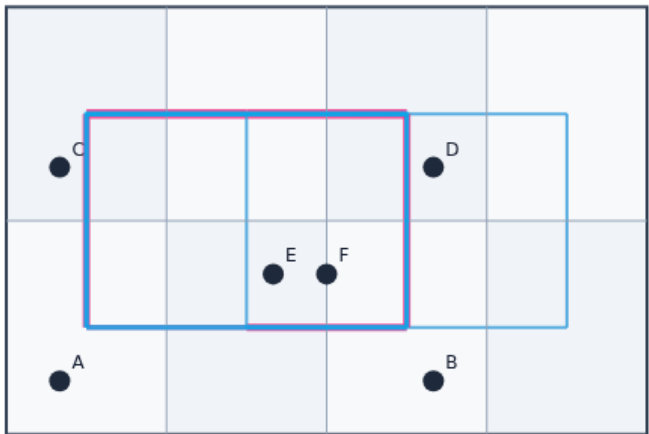
Edge cost

L overflow

MAZE ROUTING

L overflow

Step 2 / 5 · l-ov · module02-05-maze-gcell-route



Explanation

L-HV on spread overflows on shared corridors.

- Pattern first
- Then maze

total=2

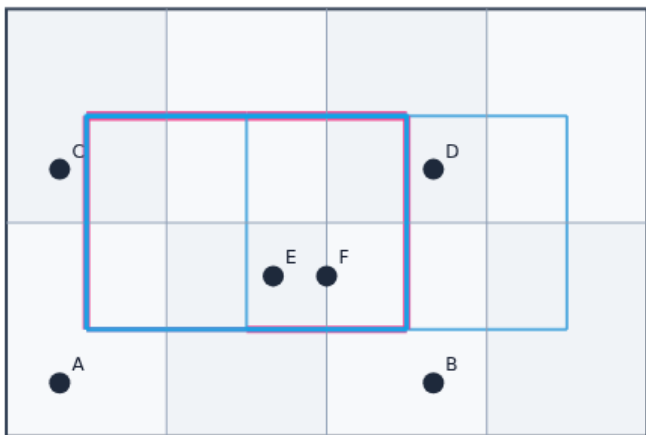
L overflow

Route maze

MAZE ROUTING

Route maze

Step 3 / 5 · maze · module02-05-maze-gcell-route



Explanation

Sequential maze routing considers current usage.

- Six nets
- Finite cost

maze=2

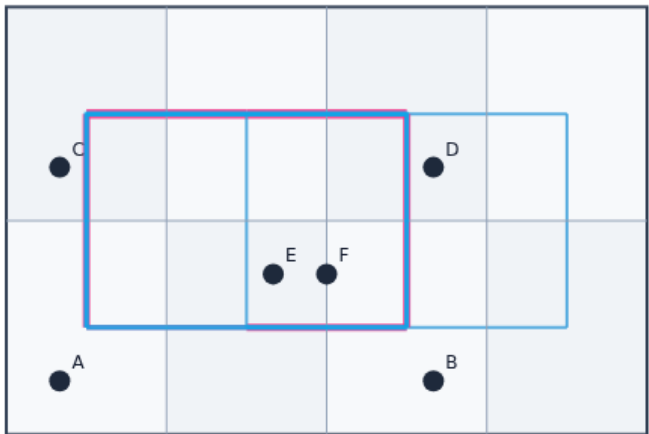
Route maze

Compare totals

MAZE ROUTING

Compare totals

Step 4 / 5 · compare · module02-05-maze-gcell-route



Explanation

Maze may match or beat L overflow on the toy.

- Not magic
- Still Cap=2

Compare L vs maze

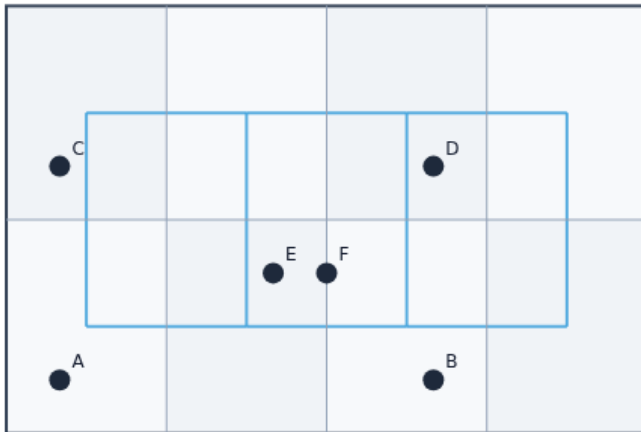
Compare totals

When to maze

MAZE ROUTING

When to maze

Step 5 / 5 · use · module02-05-maze-gcell-route



Explanation

Use maze when pattern routes saturate edges.

- Global stage
- Rip-up next

Next modules

When to maze

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Maze GCell routing

INTERACTIVE TOOL

Maze GCell routing

Maze search detours around saturated edges when L-routes overflow.

Your job: route L-HV first (overflow ≈ 2), then try Route maze. Maze may reduce or redistribute overflow.

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] L-HV has overflow

Pick a challenge and click **Start**. Move cells, route, then **Check**. Reveal golden is study-only.

StartShow hintCheckNext challengeStop checkingIdle

A B C D E F ← → ↑ ↓Reset to starterReveal golden (study)Route L-HV

Route L-VHRoute mazeRip-up reroute

Graph

Reference / metrics

selected: A @ GCell (0,0)
routes: (none — click Route)
overflow total: 0.00 · max: 0.00 · count: 0
edge capacity: 2
HPWL: 34.0
view: your placement

Browser lab starter

Implement track

- Implement `maze_route(a, b, usage, capacity, nx, ny)`
- Block edge $((0,0),(1,0))$ at cap two and show maze cannot go direct

Pitfalls

- Treating usage as per-tile instead of per-edge
- Forgetting BFS needs visited set on GCells not edges
- Returning a path through a full edge because you checked the wrong direction key

Your turn

- Pass maze goldens
- Next: connect more than two pins with a star tree